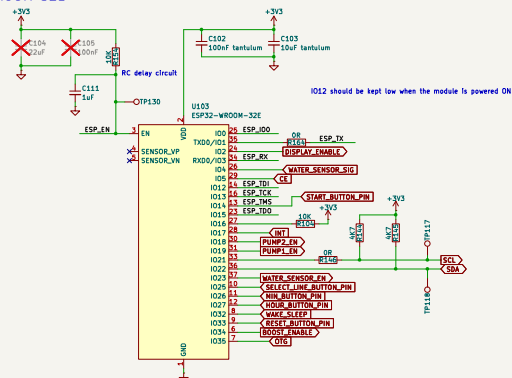


[illegible]

TO DO:

- Find suitable LED's REG / GREEN etc
- Fix the MOSFET pinout
- Find suitable capacitors (correct foot print and voltage rating)
- Re-Check schematic for correct values and symbols
- Check and Re-Do PDN block diagram and check logic
- Check the BMU thermistor and resistor values
- Re-evaluate the sleep/wake logic between RTC / ESP32 / BQ25892 QON pin
- Check bootpints for all components
- Re-check pinouts from ESP32 (libernet was not capable of hardware Interrupt)
- Re-check USB-PD Vmin and Vmax options

[illegible]

A circuit diagram of a USB Type-C connector. It shows a green wire connected to a red wire, which is then connected to a red wire. A capacitor labeled C118 with a value of 100mF is connected between the red wire and ground. The red wire is labeled VBUS.

Output Voltage:
R2 should be kept under 150K as per datasheet
 $R1 = (V_{out}/V_{ref} - 1) * R2$
 $R1 = (14V/0.8V - 1) * 121K = 2M$
R1 = 2M
R2 = 121K

Parameters:
 $V_{in} = 3V$ used as worst case according to datasheet
 $V_{out} = 5V$
 V_{ref} voltage of 0.8V as per datasheet
Inductor Selection
 $IL(DC) = (V_{out}(I_{out}) / (V_{in} \times \text{efficiency}))$
 $IL(DC) = (34W / 1A) / (3V \times 80\%) = 5.83A$
 $IL(DC) = 5.83A$
Delta IL = usually $30\% - 40\% \times IL(DC)$
 $\Delta IL = 0.4A \times 5.83A$
 $\Delta IL = 2.33A$
 $L = \Delta t_{ripple} \times [(V_{in}(V_{out} + 0.8V - V_{in})) / (f_{sw} \times V_{out})]$
 $L = 2.33A / [3V \times (1 + 0.8V - 3V)] / [600KHz \times (1$
 $L = 9.26\mu H$

The schematic diagram illustrates the USB interface circuit. At the top, a microcontroller (Q113, DMP3013SFV) is connected to a USB-to-UART bridge IC (CYPD3177-24LG). The bridge IC has pins for USB_CCC_P2, USB_CCC_N2, SDA, SCL, HPL_SDA, HPL_SCL, IFCNT, GPIO_1, FAULT, FLIP, and various control signals like ISINK_COARSE, ISINK_FINE, IBUS_MAX, IBUS_MIN, VBUS_MAX, and VBUS_MIN. These pins are connected to a series of capacitors (C159-C167) and resistors (R159-R167). Two RS-485 transceivers (Q114, DMP3035FV and Q115, 2N7002K) are shown, each with its own set of capacitors (C168-C170) and resistors (R168-R170). The circuit is powered by +VDD and +VSS.

[illegible]

The diagrams show the following connections:

- System WakeUp:** Pin 1 (VCC) connected to a 10k resistor and JST-B20 pin 114.
- Line Select:** Pin 2 (VCC) connected to a 10k resistor and JST-B20 pin 106.
- Manual Start:** Pin 3 (VCC) connected to a 10k resistor and JST-B20 pin 107.
- Hour:** Pin 4 (VCC) connected to a 10k resistor and JST-B20 pin 108.
- Minute:** Pin 5 (VCC) connected to a 10k resistor and JST-B20 pin 109.
- System Reset:** Pin 6 (VCC) connected to a 10k resistor and JST-B20 pin 111.